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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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Programme: **B.E / B.Tech**

Slot : D

Course Code: **CSA1519**

Faculty : Dr.Arul Raj A.M

Course Name: **Cloud Computing and Big Data Analytics**

Date: 03/11/2026

Duration: **1:30 Hr**

Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Overall Marks Distribution :

Question No	Question	CO	Bloom's Level	Marks	
1	Q1 – Cloud Evolution and Service Models	CO1	BL3	10	
2	Q2 – Server Consolidation Energy Saving	CO1	BL3	10	
3	Q3 – VM Over-Commitme Security	CO2	BL4	10	
4	Q4 – Data Centre, SDDC, and Privacy	CO2	BL4	10	
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10	
Total				50	

Student Signature	Faculty Signature	Program Director

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices. **(10) (CO1) (BL3)**

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks

associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

(10) (CO2) (BL3)

Given data

Parameter	Value
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

CIA - 1

Developments in computing Hardware:

High performance processors:

Modern CPUs provide multiple cores and improved processing power.

Hardware virtualization:

Technologies such as Intel VT-x and AMD-V enable multiple virtual machines to run a single physical server.

Large capacity memory:

Increased RAM capacity supports ~~via~~ multiple virtual machines to run on a single physical server. and applications.

High speed ~~storage~~ networking

Fibre optic networks and high speed internet enable remote access to cloud services.

High speed storage.

SSDs and distributed storage systems improve application performance.

Developments in Internet software.

* web browser and web applications
* REST APIs and web services.

* Distributed application architecture.
* Containerization and orchestration.
* Automated resource provisioning.
* Multi tenant application design.

Requirements	cloud model	Justification
Have existing content management applications to VMs while retaining OS control	Iaas	provide virtual machines, storage and networking with os level control
Develop and deploy a new customer facing application without maintaining servers or runtime	Paas	provides managed runtime middleware and deployment environment.
Subscribe to ready made browser	Saas	provide complete software.

Infrastructure as a Service.

* virtual machines
* Storage
* Networking
* virtual firewalls.

The customer manages the OS applications and installed software.

EX: Amazon EC2, Microsoft Azure.

Platform as a service (PaaS)

* Operating systems
* Runtime environments
* middleware
* Infrastructure
* Scaling and patching

the customer focuses on application code.

EX: Google App Engine, Azure App service.

Software as a Service (SaaS)

The provider manages the application infrastructure updates and maintenance.

EX: Salesforce CRM, Microsoft Dynamics 365.

Broader XaaS model.

STaaS - Storage as a Service
DBaaS - Database as a service
Faas - Function as a service
Naas - Network as a service
SECaas - security as a service
AIaas - Artificial intelligence as a service.

2. Server consolidation and Energy saving.

Given data

Parameter	value
Existing physical servers	15
Average CPU utilization	25%.
Power consumption per server	300 W
New virtualization host power	450 W
Target CPU utilization	65%.
N+1 fault tolerance	1
Electricity cost	2 kWh.

6

Number of active host and total host.

total workload $W = 15 \times 25\%$

$W = 3.75$ server-equilants

At 65% target utilization:

$$N = \frac{3.75}{0.65}$$

$N = 5.77$

$N = 6$ active hosts

including N+1:

$N_{total} = 6+1$

$N_{total} = 7$ hosts

b) consolidation ratio:

$$CR = \frac{\text{Original server}}{\text{New hosts}}$$

$CR = \frac{15}{6}$ $CR = 2.5:1$ //

Including the additional standby host

$CR = \frac{15}{7} = 2.14:1$ //

Active host consolidation ratio = 2.5:1

total host consolidation ratio = 2.14:1

c) CPU utilization

Total workload = 3.75 server - equivalents

when all 7 hosts.

$$U = \frac{3.75}{7} \times 100$$

$$U = 53.57\%$$

when one host fails..

$$U = \frac{3.75}{6} \times 100$$

$$U = 62.5\%$$

d) Annual Electricity cost Saving

Existing power:

$$P_{old} = 15 \times 350$$

$$= 5.25 \text{ KW}$$

Annual energy:

$$E_{old} = 5.25 \times 8760$$

$$E_{old} = 45990 \text{ kwh}$$

Annual cost:

$$Cold = 45990 \times 9$$

$$Cold = 413\,910$$

Equivalents

New power System:

$P_{new} = 7 \times 450$

$= 3.15\ kW$

Annual Energy

$E_{new} = 3.15 \times 8760$

$= 27594\ kWh$

Annual cost

$C_{new} = 27594 \times 9$

$C_{new} = 248346$

Annual saving:

$S = 413910 - 248346$

$S = 165564.$

percentage saving:

$\frac{165564}{413910} \times 100 = \boxed{40\%.}$

Q3. virtual CPU and memory over commitment Analysis:

a) CPU and memory over commitment ratios.

total allocated vCPU : $12 \times 3 = 36$

physical CPU cores = 24

$R_{CPU} = \frac{36}{24}$ $\boxed{R_{CPU} = 1.5:1}$

Memory over commitment

total allocated VM memory:

12 × 8 = 96 GB

Available memory = 88 GB

$R_{memory} = \frac{96}{88}$

$R_{memory} = 1.09 : 1$

b) Average and peak memory sufficiency:

Total memory usage:

$M_{normal} = 12 \times 5$

$M_{normal} = 60$ GB

Available RAM = 88 GB

Memory headroom : 88 − 60 = 28 GB.

peak operation:

Eight VM use 8 GB each:

8 × 8 = 64 GB

4 VM use 6 GB each:

4 × 6 = 24 GB

total = 64 + 24 ⇒ M_{peak} = 88 GB.

Average and Peak CPU sufficiency

Normal operation:

Each VM uses 35% of 3 VCPU

$CPU_{VM} = 3 \times 0.35$

$CPU_{VM} = 1.05$

For 12 VMs:

$CPU_{normal} = 12 \times 1.05$

$CPU_{normal} = 12.6$

CPU utilization:

$U = \frac{12.6}{24} \times 100$

$U = 52.5\%$

Peak operation:

Each VM use 75% of 3 VCPUs.

$CPU_{VM} = 3 \times 0.75$

$CPU_{VM} = 2.25$

For 12 VM.

$CPU_{peak} = 12 \times 2.25$

$= 27$ cores

$U = \frac{27}{24} \times 100$

$U = 112.5\%$

* Add more physical CPU cores.
* Use CPU reservation and resource limits
* Reduce unnecessary vCPU allocation
* Increase physical RAM
* Patch and Harden hypervisor
* Use role-based actions control and RBAC.
* Monitor CPU ready time, memory pressure and VM performance.

4. Cloud federation Between 2 universities

conventional cloud Data center Architecture.

* Physical server
* Storage systems
* Network Infrastructure
* Hypervisor
* cloud management software.

University A:

provides research-computing resources.

University B:

resource Limits
location

Example

* university B requests research resources.
* university A service exposes research computing
* The service authenticates the request
* computing resources are allocated to university B.

Federation challenges:

* Resource sharing
* service availability
* identity management
* Data privacy

Privacy Risks of user presence information

* tracking Researches
* Revealing work schedules
* Monitoring user activity
* unauthorized users.

Safeguard

* Obtain consent
* Restrict users
* Apply retention policies.

provides access to research resource from university A and vice versa.

limitations:

1. Across separate management systems.
2. Limited automation
3. Complex resource monitoring
4. Difficult across cloud monitoring.

SDDC Architecture:

SDDC virtualizes compute, storage and networking:

Benifits:

1. Dynamic resource allocation
2. Automated workload placement
3. Centralized monitoring
4. Flexible network configuration
5. Improved resource utilization.

POA & SOA:

SOA enables applications to communicate through standardized interfaces.

Edge filtering and surveillance computing

a) Daily Data volume and bandwidth without filtering.

Total clips per day

$20 \times 24 = 480$

$400 \times 150 = 72000$

Daily data volume

$D = 72000 \times 4$

$D = 288000$ MB

$D = 288$ GB / day

Average bandwidth:

$B = \frac{288 \times 1000 \times 8}{86400} \Rightarrow B = 26.67$ Mbps

b) 30 day storage cost without filtering.

Monthly data

$S = 288 \times 30$

$S = 8640$ GB

Storage cost:

$C = 8640 \times 1.5$

$C = 12,960$

* Encrypt sensitive data.

Privacy Risks of Federated Identity

Risk includes:

* identity impersonation.
* unauthorized access
* credential theft

Technical Safeguard

* MFA
* SAML or OpenID connect
* Encryption
* Audit logging.

Administrative Safeguards:

* Federation agreements
* Defined responsibilities
* Privacy policies
* Security audits
* Response procedures.

Edge filtering Results:

Daily data volume:

$D_{filtered} = 288 \times 0.25$

$D_{filtered} = 72\ GB/day$

Average Bandwidth:

$B_{filtered} = 26.67 \times 0.25$

$B_{filtered} = 6.67\ Mbps$

Monthly storage:

$S_{filtered} = 72 \times 30$

$S_{filtered} = 2160\ GB$

Storage cost

$C_{filtered} = 2160 \times 1.50$

$C_{filtered} = 3240.$

Monthly Storage saving.

$Saving = 12960 - 3240$

$= 9720$

Percentage Saving:

$\frac{9720}{12960} \times 100$

75%.

d) Monthly serverless invocation cost after filtering:

$$72000 \times 30 = 2160000$$

After filtering

$$2160000 \times 0.25 = 540000$$

Monthly serverless cost:

$$\Rightarrow 540000 \times 0.00015$$

$$\boxed{\Rightarrow \$1}$$

Role of REST API

* uploading video clips
* sending metadata
* Authentication and authorization.
* Standardized communication

Why object storage is suitable

* It stores large unstructured data
* It provides high scalability
* It supports metadata
* It supports high durability.